SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Qart.ai
Project / Solution Name	Qart.ai: AI-era reskilling for Malaysia's GBS finance workforce
Name (Leader)	Kydyrgaliyev Meiirman
Email & Phone (Leader)	meirman02.07@gmail.com · +60 11-2725 6756
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://qartai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Moo7x/qart.ai (public; Apache License 2.0)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A React web app is served by a FastAPI backend on Replit Autoscale, over PostgreSQL through one SQLAlchemy layer. A synthetic CV is scrubbed of names, IC numbers, phones and emails; OpenAI gpt-4.1-mini reads the scrubbed text into skill claims under a strict JSON schema, and code checks every quote against the CV; the same CV always gets the same stored reading. A Replit Scheduled deployment runs the evidence pipeline daily: seven public sources stored as dated, immutable observations. From them code trains a task-level AI-use model (gradient boosting on BGE-M3 features from a RunPod serverless GPU, chosen by cross-validation), backtests job-group demand forecasts and labels each role Growing, Transforming, Shrinking, Emerging or Stable. Fit, skills gap, scenario weights, ordering and the reskilling path are computed in Python; the language model never writes numbers. A catalogue job finds courses and videos on doing each skill with AI, ranks them with BGE-M3, the BGE reranker and a learned ranker, and keeps only those whose quotes are verified on the page or captions. New evidence proposes revisions that users accept or decline.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12; TypeScript 6.0	Backend, pipeline and tests; web app	PSF; Apache-2.0
FastAPI 0.155.0 (Starlette), Uvicorn 0.54.0, Pydantic 2.14.1	HTTP API, server, validation	MIT / BSD-3-Clause / MIT
SQLAlchemy 2.1.6 (Core), psycopg 3.3.6	One database layer for SQLite and PostgreSQL	MIT; LGPL-3.0 (unmodified)
scikit-learn 1.9.1, numpy 2.5.4; LightGBM 4.6.0, XGBoost 3.0.2 (optional)	Task-level AI-use model, demand forecasts, resource ranker	BSD-3-Clause; MIT; Apache-2.0
openai 3.36.1 (Python SDK)	Client for gpt-4.1-mini	Apache-2.0

Name & Version	Purpose in Project	License
httpx2 2.15.1, certifi 2026.10.23, PyYAML 6.0.3	Calls to sources and RunPod; data packs	BSD-3-Clause; MPL-2.0; MIT
pypdf 6.21.4, python-docx 1.2.0, python-multipart 0.0.32	Reading synthetic CVs (PDF, DOCX) and uploads	BSD-3-Clause; MIT; Apache-2.0
youtube-transcript-api 1.2.4, trafilatura 2.3.1	Reading captions and page text to verify resource quotes	MIT; Apache-2.0
React 19.3, react-router-dom 7.18, Recharts 3.10, lucide-react 1.61, qrcode 1.5.4	User interface, routing, charts, icons, resume QR	MIT; ISC
Vite 8.3, Tailwind CSS 4.3, oxlint 1.88	Web build, styling, lint	MIT
pytest 9.1.1, ruff 0.17.1, mypy 2.4.0, Locust 2.47.0 (dev)	452 tests, lint, types, load test	MIT
sentence-transformers 6.2.0, transformers 5.22.1, PyTorch 2.6 (RunPod worker only)	Loading BGE-M3 and the reranker on the GPU	Apache-2.0; BSD-3-Clause
Fonts: Inter, Instrument Serif, Geist Mono (Google Fonts)	Interface typography	SIL OFL 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API: gpt-4.1-mini	Reading scrubbed CVs into quoted skill claims; plain-words explanations; resource and course fit checks; the scout and progress-check agents	paid, the team's own key
RunPod Serverless API (our endpoint)	BGE-M3 embeddings and BGE reranking for job titles, tasks and resources	sponsor-provided credits
YouTube Data API v3; Microsoft Learn Catalog API	Finding videos and courses per skill (metadata and links only)	free
OpenDOSM API (api.data.gov.my); DOSM Job Market Insights pages	Malaysian employment and online job-ad evidence	free, public
Hugging Face dataset download (Anthropic Economic Index); O*NET 31.0 files; ILO GitHub	AI-use, task and exposure evidence	free, public
ESCO API v1	ISCO-08 group for each O*NET occupation (cached; app works without it)	free, public

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale deployment	Hosts the web app and API (the live prototype)	sponsor-provided credits
Replit PostgreSQL (production database)	Profiles, plans, CV readings, evidence, proposals, usage	included with the Repl

Service & Provider	Purpose in Project	Plan / Region
Replit Scheduled deployment	Runs the evidence pipeline daily into the same database	sponsor credits · 1 vCPU / 2 GiB · daily
RunPod Serverless endpoint	GPU inference for BGE-M3 and the reranker; scales to zero	sponsor credits · 16/24 GB GPU · max 1 worker
GitHub (repository and Actions)	Source code; RunPod builds the worker image from it; Actions trigger the pipeline	free, public repository

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-4.1-mini	OpenAI (API)	pre-trained, called with strict JSON schemas	OpenAI API terms
BAAI/bge-m3	Hugging Face, served on our RunPod endpoint	pre-trained (embeddings)	MIT
BAAI/bge-reranker-v2-m3	Hugging Face, served on our RunPod endpoint	pre-trained (cross-encoder)	Apache-2.0
Task-level AI-use model (gradient boosting on BGE-M3 features)	Qart.ai (src/forecast/task_model.py)	self-trained on 2,691 Economic Index tasks; 5-fold MAE 11.2 vs 16.2 baseline	ours (Apache-2.0)
Job-group demand forecasts (HistGradientBoosting)	Qart.ai (src/evidence/forecast.py)	self-trained on OpenDOSM; used only when it beats "no change" in backtest	ours (Apache-2.0)
AI-resource ranker (logistic regression)	Qart.ai (src/collectors/ai_resources.py)	self-trained on model labels with verified quotes; CV AUC 0.73	ours (Apache-2.0)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Anthropic Economic Index, release 2026-06-26	huggingface.co/datasets/Anthropic/EconomicIndex	public	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0 Database (tasks, ratings, software skills, related occupations)	onetcenter.org	public	CC BY 4.0	N
ILO WP140 GenAI exposure scores (Gmyrek et al., 2025)	github.com/pgmyrek/2025_GenAI_scores_ISCO08	public	CC BY 4.0	N
OpenDOSM wages_occupation; DOSM Job Market Insights Q2 2023 – Q2 2026	api.data.gov.my; dosm.gov.my	public	Malaysian Government Open Data Terms 1.0	N
TalentCorp GBS impact study (2024); MyCOL 2024/2025	talentcorp.com.my (published PDFs)	public (facts cited with pages)	facts only, no text reproduced	N
ESCO–O*NET crosswalk	esco.ec.europa.eu	public	European Commission reuse notice	N
Synthetic CVs and personas (data/samples)	made by the team	synthetic, labelled	Apache-2.0	N
Resource labels for the ranker (data/packs/gbs_fa)	gpt-4.1-mini judge, quotes verified in code	self-collected (model labels)	Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic, Claude Opus models)	Code, tests, documentation, data checks, design changes, research summaries, Bahasa Malaysia strings (native review pending)	major
Codex (OpenAI)	Journey verification, fixes, O*NET software mapping, tests and documentation	moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change went in small commits that we reviewed and can explain. 452 automated tests, ruff, mypy and lint ran before each commit, and screens were checked in a browser against the real API with two synthetic CVs. Inside the product the language model never produces numbers: code computes them, every quote is checked against the CV, page or captions, and every fallback is labelled. Model-labelled resources were reviewed a second time and the results are published in docs/RESOURCE_EVAL.md. Each tool, model and dataset is logged with its licence in docs/DECLARATION_LOG.md.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	credits: Autoscale and Scheduled deployments, PostgreSQL	Hosting the live prototype, its production database and the daily evidence pipeline
RunPod	credits: Serverless GPU endpoint	BGE-M3 embeddings and the BGE reranker: job-title matching, task-model features and resource ranking

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to the deployed prototype as built (web app, database, daily collection), but any Python host with PostgreSQL could replace it. RunPod carries the quality of matching and forecasting: on our 40-title test, top-1 occupation accuracy is 72% with the GPU models and 60% with keyword search, and the task-level AI-use model is built on BGE-M3 features. The app keeps working without RunPod: every call has a time budget and a labelled keyword fallback, and the trained model is stored, so no page waits for a GPU cold start.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Only synthetic data is used; sample CVs are fictional and labelled. There are no accounts. An uploaded CV is read in memory; names, IC numbers, phones and emails are removed first, and only the scrubbed text goes to OpenAI. Job titles and task text go to our RunPod endpoint. We store skill claims with their quoted CV sentence, plans, decisions and the model's reading of a CV keyed by a hash of the scrubbed text (the text itself is not kept). Protected attributes are never used. A person can export or delete everything from the Privacy page, and idle profiles and stored readings are removed after 90 days. Keys live in environment variables, never in code. Usage is counted per purpose without content. Platform totals hide groups smaller than five. The design follows Malaysia's PDPA principles (docs/PRIVACY_PDPA.md).

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Misreading: gpt-4.1-mini can misread a CV; every claim must quote the CV, code rejects quotes that are not there, and levels stay editable claims. Bias: protected attributes are removed and a paired-profile test checks identical results. Forecast limits: AI use comes from global Claude conversations, not Malaysian workplaces; exposure is potential, not job loss; DOSM demand is by broad job group, not role; each is labelled observed, assumption or estimate, and a role is called Transforming only when two of three AI signals agree. Resource labels come from a model, not people (second review: 32 of 41 clearly right). Provider dependency: RunPod cold starts and OpenAI limits have time budgets and labelled fallbacks. Scope: GBS finance only.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Kydyrgaliyev Meiirman (team leader)

1 November 2026